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DUNMAN HIGH SCHOOL **Preliminary Examination** **Year 6**

H2 BIOLOGY

9648/03

Paper 3 Applications Paper and SPA Planning Task (Booklet I)

20 Sep 2011

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A:

Consists of 5 Structured Questions separated into Booklets I and II.

Answer **all** questions.

Write your answers in the **space provided** on the question paper.

Section B:

Consists of 1 SPA Planning Task found in Booklet II.

Write your answers on the separate **writing papers** provided.

At the end of the examination, fasten all your work securely together.

Section C:

Consists of 1 Essay Question found in Booklet II.

Write your answers on the separate **writing papers** provided.

At the end of the examination, fasten all your work securely together.

Sections A, B and C are to be submitted **separately**.

The number of marks is given in brackets [] at the end of each question or part question.

Do not open the booklets until you are told to do so.

BOOKLET I		
FOR EXAMINER'S USE		
Section A1	1	/7
	2	/7
	3	/6

This document consists of **6** printed pages and **2** blank pages.

Booklet I

Section A1: Structured Questions (20 marks)

Answer **all** questions in this section.

Question 1

For
Examiner's
use

- (a) Chymosin is an enzyme used to coagulate milk in cheese-making. It was obtained from the stomachs of calves. Now it can be made using gene technology.

The scientists who first used gene technology to produce this enzyme inserted the chymosin gene into plasmids. The plasmids were then mixed with yeast cells.

Some of the yeast cells took up the plasmids and some did not. The yeast cells which took up the plasmids produced chymosin.

- (i) What is a marker gene? [2]

- (ii) A marker gene could be used to obtain a sample of yeast cells which all contain the chymosin gene. Explain how this is possible. [2]

- (iii) Antibiotic resistance genes are often used as marker genes. Explain why some people are concerned about the use of antibiotic resistance genes. [2]

- (iv) Suggest **ONE** advantage of making cheese with chymosin from yeast, rather than from calves. [1]

Total: [7]

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Question 2

Fig. 2.1 below shows the isolation of a specific human gene and cloning into a bacteriophage.

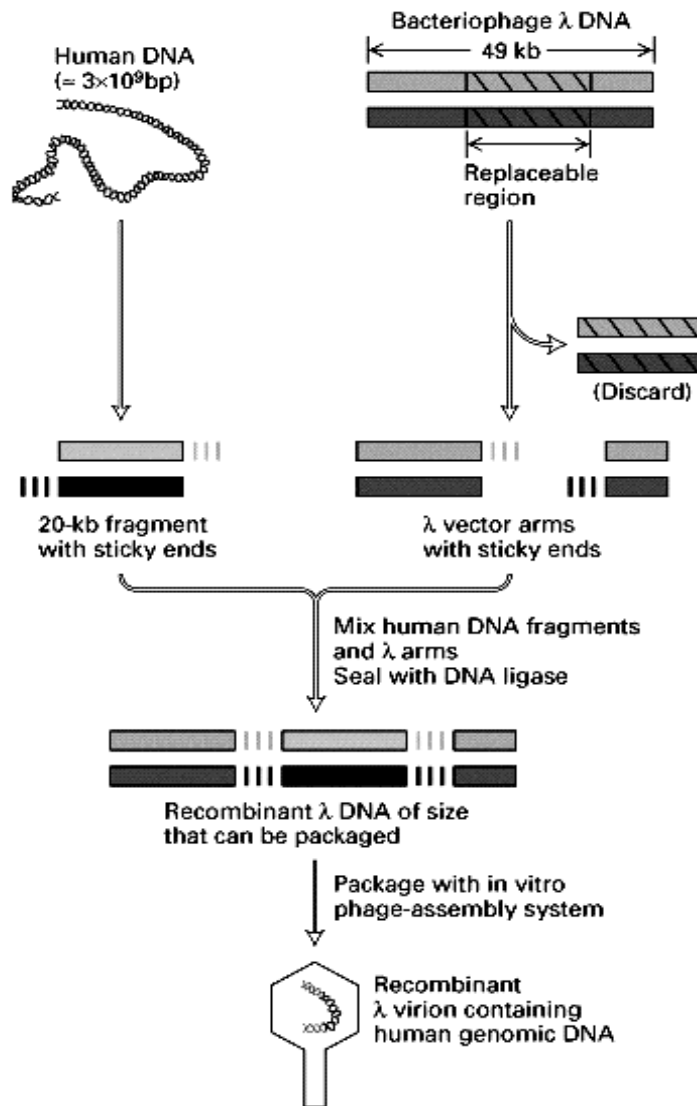


Fig. 2.1

(a) Identify the type of DNA library derived from this method. [1]

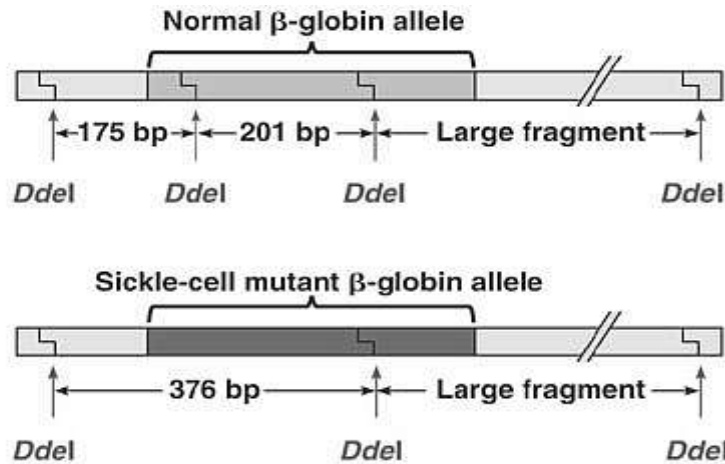
- (b) Name another vector that can be used for the making of DNA library and give **TWO** reasons why this vector might be favoured over a bacteriophage vector. [3]

- (c) Describe **THREE** differences in the use of cDNA library and genomic library. [3]

Total: [7]

Question 3

- (a) **Fig. 3.1** below shows the positions of the restriction sites for a restriction enzyme, *Ddel* along the normal hemoglobin beta chain gene and the mutant hemoglobin beta chain gene.

**Fig. 3.1**

State the number of fragments that can be obtained from the normal hemoglobin beta chain gene and the mutant hemoglobin beta chain gene after cutting using *Ddel*. Explain your answers. [4]

- (b) When the normal Beta A gene is digested with DdeI and the fragments separated by electrophoresis, the probe bind to a short fragment (length indicated by the two arrows). However, the enzyme cannot cut the mutant Beta S gene at this site, so the probe attaches to a much larger fragment (length indicated by the two arrows).

The pedigree below shows a family whose only son has sickle-cell disease. Both his father and mother were heterozygous. The electrophoresis patterns for both parents and child 2 are placed directly beneath them.

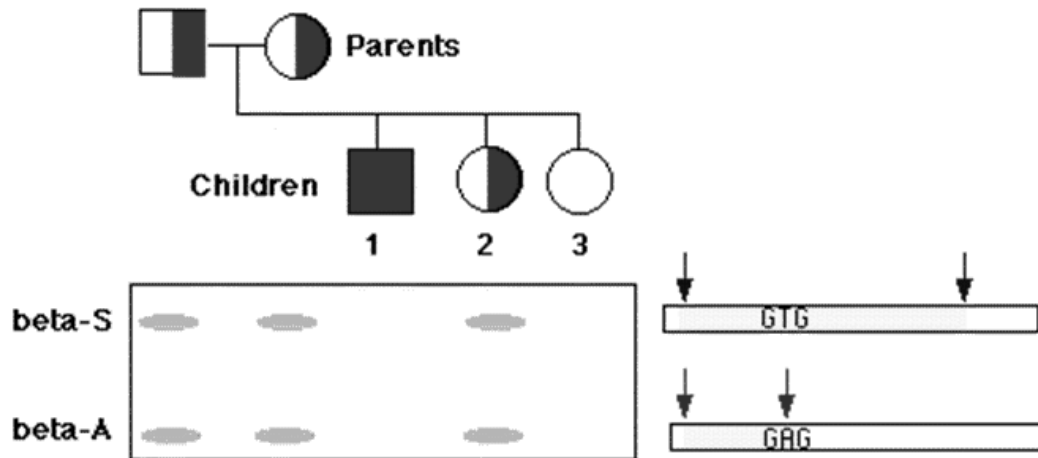


Fig. 3.2

Draw on **Fig. 3.2** to show how the electrophoresis pattern/(s) for child 1 and child 3 would look like. [2]

Total: [6]

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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY

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Paper 3 Applications Paper and SPA Planning (Booklet II)

20 Sep 2011

2 hours

Additional Materials: Writing paper

BOOKLET II		
FOR EXAMINER'S USE		
Section A2	4	/10
	5	/10
Section B	6	/12
Section C	7	/20

This document consists of **8** printed pages and **2** blank pages.

Booklet II

Section A2: Structured Questions (20 marks)

Answer **all** questions in this section.

*For
Examiner's
use*

Question 4

Micropropagation can be defined as growing plants from small pieces of tissue under **sterile conditions** in a laboratory on specially selected media. The tissues first form calli, after which they develop into plantlets. The plantlets formed are said to be clones.

This is performed in a carefully controlled environment. The media that the plants are grown on contain a carbohydrate source, a range of mineral salts and agar. Sometimes vitamins, amino acids, **plant growth regulators** or plant extracts may be added to encourage growth.

- (a) Explain why it is important for micropropagation to be conducted under sterile conditions. [1]

- (b) Explain why an infection is more likely to destroy a complete batch of plants grown by micropropagation than a batch of plants grown from seeds. [3]

- (c) Other than the high losses if contamination occurs mentioned in (b), explain **TWO** limitations of micropropagation. [2]

- (d) Equal masses of tobacco plant callus were cultured for four weeks on media containing different concentrations of two plant growth regulators, auxin and cytokinin.

The results are shown in **Table 4.1**.

Table 4.1

Treatment	Concentration of plant growth regulators / mg dm ⁻³		Effect of plant growth regulators on callus growth
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

With reference to **Table 4.1**,

- (i) Describe the effects of auxin and cytokinin on callus growth. [3]

- (ii) State which treatment (**A** to **E**) should be given to the shoots developed from callus in treatment **D** to turn them into plantlets. [1]

Total: [10]

Question 5

- (a) Salmon growth hormone (sGH) has been inserted into plasmids and the recombinant plasmids were microinjected into eggs from very slow-growing domestic and wild strains of Atlantic salmon.

Their growths were scored, as shown in **Fig. 5.1**. In a further analysis, growth hormone was extracted from domestic-GH treated salmon on dates 11/5, 23/6, 14/7 and 1/9. A protein gel was run and the results shown in **Fig. 5.2**.

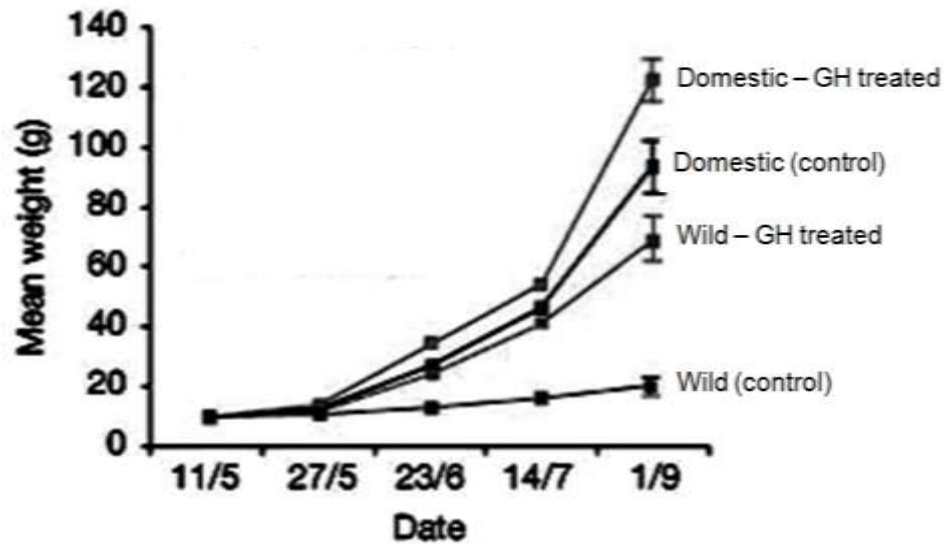


Fig. 5.1



Fig. 5.2

Using the information given,

- (i) describe the effect of the growth hormone on salmon. [2]

- (ii) suggest why there is a difference in weight gain between domestic-GH treated and wild-GH treated fish. [1]

- (iii) state and explain the process sGH undergoes before exerting its effect on weight gain. [2]

- (b) In 1993, the Food and Agriculture Organisation (FAO) of the United Nations published the Code of Practice on the Introduction and Transfer of Marine Organisms. It included the following recommended procedures for the consideration of the release of genetically modified organisms (GMOs):

(a) Recognizing that little information exists on the genetic, ecological, and other effects of the release of genetically modified organisms into the natural environment (where such releases may result in the mixing of altered and wild populations of the same species, and in changes to the environment), the Council urges Member Countries to establish strong legal measures to regulate such releases, including the mandatory licensing of physical or juridical persons engaged in genetically modifying, or in importing, using, or releasing any genetically modified organism.

*(b) Member Countries contemplating any release of genetically modified organisms into open marine and fresh water environments are requested at an early stage to notify the Council before such releases are made. This notification should include a risk assessment of the **effects of this release on the environment and on natural populations**.*

*(c) It is recommended that, whenever feasible, **initial releases of GMOs be reproductively sterile** in order to minimize impacts on the genetic structure of natural populations.*

(d) Research should be undertaken to evaluate the ecological effects of the release of GMOs.

Source: <http://www.fao.org/DOCREP/003/W3592E/w3592e0b.htm>

- (i) Explain how releasing reproductively sterile GMOs would help to prevent negative consequences on the environment. [1]

- (ii) Using the example of transgenic salmon, suggest and explain **TWO** ways that the release of reproductively sterile GMOs in open seas could *still* affect the environment and natural populations of salmon. [4]

Total: [10]

Section B: SPA Planning Task (12 marks)
Write your answers on the writing paper provided.
A **NIL RETURN** is required.

Question 6

The koalas are endemic to Australia. On an expedition to Madagascar, a biologist observed some tree climbers which strongly resemble koala bears in the Madagascan forest. These tree climbers are twice the size of a normal koala.

Some members of the expedition team thought that these tree climbers may be a new species while the others thought that they are descendents of the 'giant' koala' which exists in fossils dating back more than 40,000 years.

Plan an investigation to investigate if these large tree climbers found in Madagascar are descendents of the 'giant' koala'.

Your planning must be based on the assumption that you have been provided with the following apparatus and materials.

1. tissue samples from museum specimens and the Madagascan tree climbers under investigation
2. pestle and mortar
3. DNA extraction buffer solution
4. glass rod
5. microcentrifuge tubes
6. centrifuge
7. restriction enzyme
8. agarose or polyacrylamide gel plate
9. suitable source of electrical current
10. radioactive probe
11. nitrocellulose membrane
12. autoradiography equipment

Your plan should have a clear and helpful structure to include:

- an explanation of the theory to support your practical procedure
- a description of the method used including the scientific reasoning behind the method,
- the type of data generated by the experiment
- how the results will be analysed including how the origin of the organism can be determined

[Total: 12]

Section C: Essay Question (20 marks)

Write your answers on the writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL RETURN** is required.

Question 7

- (a)** Describe and explain the use of RFLP analysis with reference to the analysis of DNA molecules. [6]
- (b)** Compare and contrast zygotic, embryonic and adult stem cells. [6]
- (c)** Discuss the social and ethical considerations for and against the use of gene therapy. [8]

[Total: 20]

END OF PAPER